
(a)
more than 50,000 gates: strongest deterministic seed, no swarm (large mode)
Mapped input C (Qiskit O3, IBM Eagle) $\{rz, sx, x, ecr\}$
Phase 1: swarm over the decisions of a sound compiler (25% of T)
Phase 2a: memetic resynthesis of the $\leq$3 elites ($k \leq 4$)
Phase 2b: deepening of the best circuit until T ($k \leq 5$)
Native re-synthesis: CX $\rightarrow$ ECR in native direction, fewest rz/sx/x
Output C^*: same qubits, couplings and ECR directions
decoder: variant lowering $\rightarrow$ pass schedule (cancel, fold, hred, kak, tket) $\rightarrow$ clean-up $\rightarrow$ native gates; every candidate is exact
CNOTs deleted only inside each block's own skeleton; run-time checks: block re-simulation (10^{-8}), round check with random states (10^{-7})
(b)
q_0
q_1
q_2
u
U3
rz
(i) convex block B of the mapped circuit (at most k qubits)
(ii) template $V_S(\mathbf{x}) = L_m \cdots L_1 L_0$ on the block's own CNOT skeleton S
(iii) delete CX_j, re-fit $\mathbf{x}$ by Levenberg–Marquardt; accept if Eq. (6) holds
(iv) Z-slot sparsification: U3 slots become free rz where re-instantiation succeeds